
Online Movie Booking System

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System Design

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1.

Customer Problem Statements & System Requirements

Customer Problem Statement:

"As someone who loves going to cinemas to view movies on the big screen, it is often a hassle and quite frustrating to book tickets, both in-person and online. Never mind not understanding how to operate the dinosaur kiosk-machinery they have on-location, but sometimes the line to use that technology is so long! Not to mention seat availability never seems to match up- employees tell me one thing, their online website tells me another, and the on-site kiosks seem to have a mind of their own. And to get a refund? Too confusing! I think a new and improved software system could help solve some of these issues by first and foremost organizing movies available, theatres available, movie times available, and seats available in a more intuitive manner, like an overview view that condenses instead of having to scroll through a plethora of options. By starting big then narrowing down the parts, we moviegoers will have a quicker and more efficient time browsing our options and purchasing the tickets. I would also expect a way for rewards program members to obtain their loyalty rewards, such as promotions or discounts, automatically- as trying to do so myself is yet another step in the already confusing path to booking movie tickets."

Glossary of Terms:

Term	Definition
Showtime	The scheduled time when a specific movie is played at a specific theatre location
Seat Selection	The process of choosing a specific seat for a movie from a visual grid layout of the theatre 
Booking ID	A unique identifier assigned to each confirmed ticket reservation
Payment Gateway	A secure interface that processes online payments via physical card or card in a digital wallet
User Profile	A personalized user account storing

	unique preferences (for recommendations), purchase history, card details, and contact info
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System Requirements:

Functional Requirements-

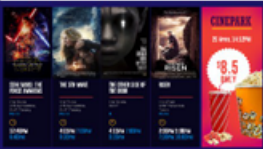
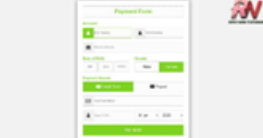
No.	Priority Weight	Description
REQ-1	High	Users should be able to browse available movies and showtimes
REQ-2	High	Users should be able to select seats from a visual layout
REQ-3	High	The system should allow secure online payments
REQ-4	Medium	Users should be able to create and manage personal profiles
REQ-5	High	The system should generate unique booking ID's and confirmation messages
REQ-6	Medium	Users should be able to cancel or modify bookings before showtimes
REQ-7	Low	The system should offer movie recommendations based on user purchase history
REQ-8	Medium	Theatre managers should be able to add, remove, or edit movies/movie details
REQ-9	High	The system should prevent double-booking of seats
REQ-10	Medium	Users should receive email or text message reminders before showtimes

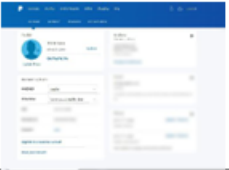
Nonfunctional Requirements-

Category	Priority	Description
Functionality	High	The system should run all key features without errors

Usability	High	Interface should be intuitive, mobile-friendly, and accessible
Reliability	High	System should handle peak traffic smoothly, without crashing or lagging
Performance	Medium	Pages should load within seconds, booking confirmation sent immediately after purchase
Supportability	Medium	System should allow for easy real-time updates and maintenance without going down

User Interface Requirements:

UI Component	Priority	Description
Login/Signup Page 	High	Users should be able to register or log in with email/social accounts
Movie Listing Page 	High	Should display movies with posters, ratings, descriptive blurbs, and showtimes
Seat Selection Page 	High	An interactive layout showing available and booked seats via color coding with an attached key
	High	A secure form for entering payment details, such as card number and CVV

Booking Confirmation 	High	A page to display user booking ID, movie details (location and time), and seat information (seat number and row in what theatre room)
Profile Dashboard 	Medium	A user hub to view past bookings, update preferences, and show membership/loyalty/rewards statuses

Plan of Work:

Task	Status	Notes
Define Project Scope	Completed	Based on initial project proposal guidelines-still awaiting feedback and grading
Gather system requirements	Completed	Functional and nonfunction requirements outlined- I believe I have downloaded all the required software programs I need as well as test them to make sure they are connected correctly and running accurately
Design UI sketches	In Progress	Working on layout ideas for all the principal main pages
Set up development environment	In Progress	Tools and frameworks to be selected as well as begin writing up the basic foundational code
Build initial prototype	Upcoming	Will start after UI and backend structure are

		complete and finalized- putting all the parts together
Testing and feedback collection	Upcoming	Planned for post-prototype

2.

Functional Requirements Specification

Stakeholders:

- Customers: End users who browse movies, book tickets, and make payments.
- Cinema Managers: Oversee movie schedules, seat availability, and revenue tracking.
- System Administrators: Support the platform, manage user accounts, and troubleshoot issues.
- Sponsors/Owners: Fund the system and check performance metrics.
- Developers: Build and sustain the system's features and infrastructure.

Primary Actors:

- Customer: This actor can browse movies and showtimes, select seats and book tickets, and make payments and receive confirmation messages.
- Cinema Manager: The cinema manager can add/update movie listings and showtimes, as well as monitor bookings and manage seat availability.

Secondary Actors:

- System Administrators: This can manage user accounts and system configurations, as well as handle escalated issues and keep data integrity.
- Payment Gateway: This can process transactions securely via credit card or digital wallet.

Use Cases:

Customer (total: 14)

- Register/Login (2): To create an account or log in to access booking features.
- Browse Movies (2): To view current and upcoming movies with showtimes and trailers.
- Select Showtime (2): To choose a preferred date and time for a movie.
- Choose Seats (2): To pick available seats from a visual layout.
- Make Payment (2): To pay for selected tickets via credit card or wallet.
- Receive Confirmation (2): To get booking details via email or SMS.
- Cancel Booking (2): To cancel a ticket before showtime and request refund.

Cinema Manager (total: 10)

- Add Movie (2): To upload movie details including title, genre, duration, and poster.
- Schedule Showtimes (2): To dates, times, and screen assignments.

- Update Availability (2): To change seat layout or mark seats as unavailable.
- View Bookings (2): To track ticket sales and occupancy rates.
- Remove Movie (2): To remove outdated or canceled listings.

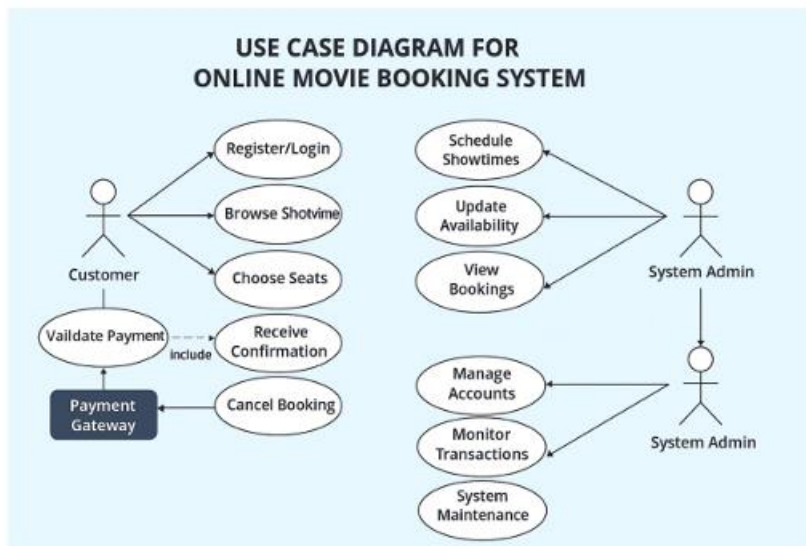
System Admin (total: 8)

- Manage Accounts (2): To add, remove, or update user profiles.
- Monitor Transactions (2): To review payment logs and resolve disputes.
- System Maintenance (2): To perform updates and fix bugs.
- Generate Reports (2): To create summaries of revenue, traffic, and usage.

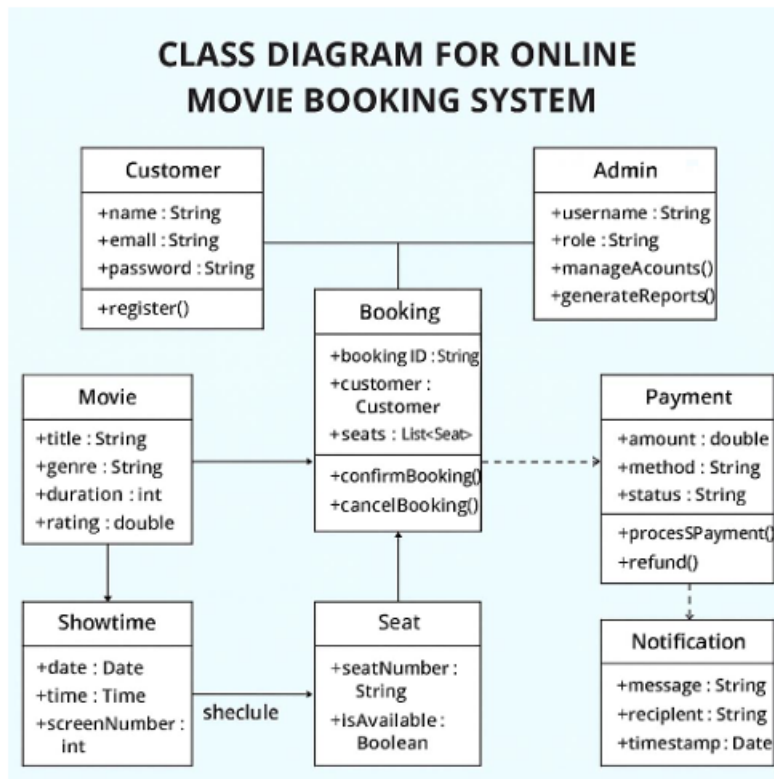
System (total: 8)

- Validate Payment (2): To confirm transaction success before issuing ticket.
- Send Notifications (2): To deliver booking confirmations and reminders.
- Update Seat Map (2): To reflect real-time seat availability.
- Log Activity (2): To record user actions for analytics and security.

Use Case Diagram:



Class Diagram:



Data Types and Operation Signatures:

-Customer-

- + name : String
- + email : String
- + password : String
- + register() : void
- + login() : boolean
- + bookTicket(movie : Movie, showtime : Showtime, seats : List<Seat>) : Booking

-Movie-

- + title : String
- + genre : String
- + duration : int
- + rating : double
- + addMovie() : void
- + updateMovie() : void

-Showtime-

- + date : Date
- + time : Time
- + screenNumber : int
- + scheduleShowtime(movie : Movie) : void
- + updateTime(newTime : Time) : void

-Seat-

- + seatNumber : String
- + isAvailable : Boolean
- + reserve() : void

+ release() : void

-Booking-

+ bookingID : String

+ customer : Customer

+ seats : List<Seat>

+ status : String

+ confirmBooking() : void

+ cancelBooking() : void

-Payment-

+ amount : double

+ method : String

+ status : String

+ processPayment() : boolean

+ refund() : boolean

-Admin-

+ username : String

+ role : String

+ manageAccounts() : void

+ generateReports() : Report

-Notification-

+ message : String

+ recipient : String

+ timestamp : Date

+ sendEmail() : void

+ sendSMS() : void

Relationship Summary:

Customer → Booking (association)

Booking → Movie, Showtime, Seat, Payment (association)

Admin → Movie, Showtime (association)

Notification → Customer (association)

Seat → Booking (association)

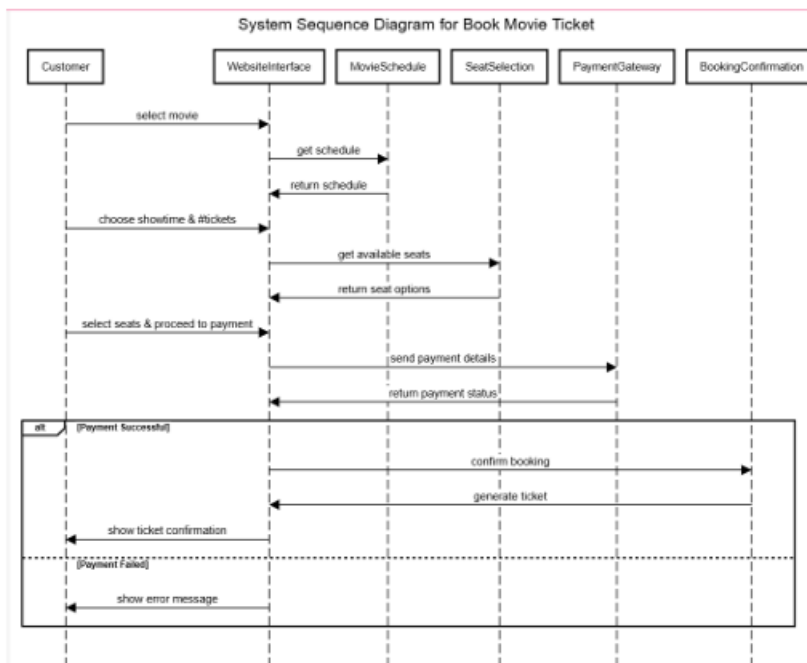
Sequence Diagrams:

Book Movie Ticket

Actor: Customer

Objects: Website Interface, Movie Schedule, Seat Selection, Payment Gateway, Booking Confirmation

1. Customer signs into the website and selects a movie
2. Website interface retrieves selected movie schedule from Movie Schedule
3. Customer picks desired showtime and number of tickets
4. Website interface requests available seats from Seat Selection
5. Customer chooses open seats and proceeds to payment
6. Website interface sends payment information to Payment Gateway
7. Payment Gateway processes payment and returns verification status
8. If successful, website interface confirms booking with Booking Confirmation. Which generates a ticket to send to customer
9. If unsuccessful, website interface displays error message



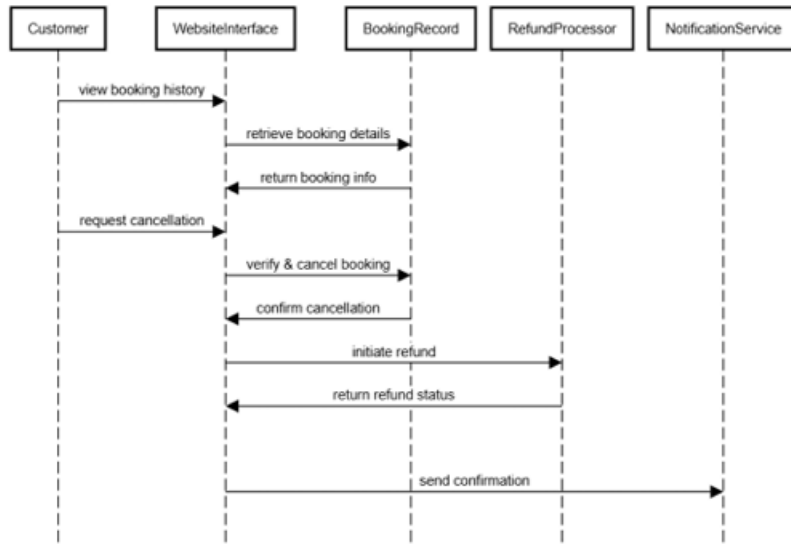
Booking Cancellation and Refund

Actor: Customer

Objects: Website Interface, Booking Record, Refund Processor, Notification Service

1. Customer signs into the website and goes to their purchase history
2. Website interface retrieves booking details from Booking Record
3. Customer chooses desired booking and requests to cancel
4. Website interface sends cancellation request to Booking Record
5. Booking Record verifies eligibility and confirms the cancellation request
6. Website interface delivers refund request to Refund Processor
7. Refund Processor processes the refund and returns verification status
8. If successful, Notification Service delivers confirmation to customer
9. If unsuccessful, website interface displays error message

System Sequence Diagram for Booking Cancellation and Refund



Activity Diagrams:

Book Movie Ticket

States

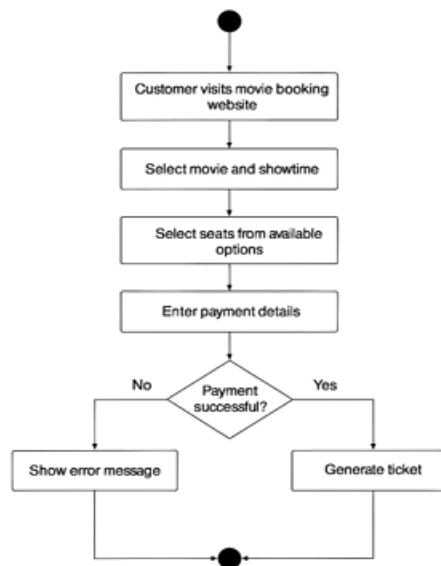
Initial State: Customer visits the online movie booking website

Final State: 1. Customer successfully receives digital movie ticket 2. Payment is unsuccessful and customer is shown an error message

Actions

The customer begins by logging into the system and selecting a movie. The customer selects a showtime and number of tickets. The system displays open seats for the desired showtime. The customer selects specific available seats and proceeds to payment portal. The customer enters their payment details. The system processes the payment. If successful, the booking is confirmed by the system and a digital ticket is sent to customer. If unsuccessful, the system displays an error message.

Activity Diagram for Book Movie Ticket



Activity Diagram for Book Movie Ticket

Booking Cancellation and Refund

States

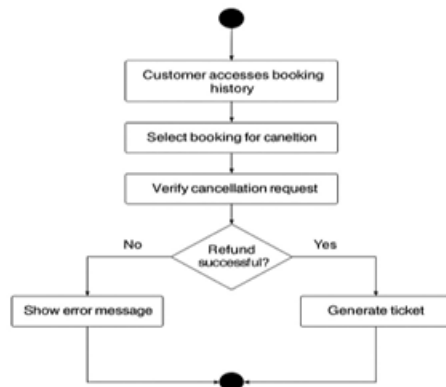
Initial State: Customer pulls up booking history

Final State: 1. Customer receives confirmation that the refund was successful 2. Refund request is unsuccessful, and system displays an error message

Actions

Customer signs into website and accesses their purchase history/ Customer selects booking for cancellation. System verifies request and eligibility. If eligible, system begins refund process. Refund processor returns status. If successful, system sends confirmation notification. If unsuccessful, system displays an error message.

Activity Diagram for Booking Cancellation and Refund



Activity Diagram for Booking Cancellation and Refund

User Interface Specification

Search for Movies:

1. Input Elements

- Search bar (text input)
- Genre dropdown
- Date choice
- Location selector
- "Search" button

2. Results Display

- Grid of movie posters with the title, rating, and "View Details" button

3. Navigation Path

- Home -> Search bar -> Filter options -> Search -> Results page

The mockup shows the 'Search for Movies' interface. At the top is a search bar with a magnifying glass icon and placeholder text 'Search for movies...'. Below it are three filter sections: 'GENRE' with a 'Select Genre' dropdown, 'DATE' with a 'Select Date' dropdown, and 'LOCATION' with a 'Select Location' dropdown. A red 'SEARCH' button is centered below these filters. The 'SEARCH RESULTS' section displays a grid of three placeholder cards. Each card contains a placeholder image, the title 'Title', a rating '★ Rating', and a 'VIEW DETAILS' button. The bottom of the screen features a navigation bar with three icons.

View Movie Details

1. Input Elements

- Click "View Details" on desired movie card

2. Results Display

- Movie poster, synopsis, cast, runtime, ratings
- Showtimes dropdown
- "Book Now" button

3. Navigation Path

- Search results -> View details -> Movie info page

The mockup shows the 'View Movie Details' interface. It features a large placeholder for a movie poster on the left. To the right of the poster, the movie's 'Title' is displayed, followed by a list of details: 'Synopsis', 'Cast', 'Runtime', and 'Rating'. Below this information is a 'Showtime' section with a 'Select Showtime' dropdown menu. At the bottom is a red 'BOOK NOW' button. The bottom of the screen features a navigation bar with three icons.

Book Tickets

1. Input Elements

- Showtime dropdown
- Seat map (clickable seats)
- Quantity selector
- "Confirm booking" button

2. Results Display

- Booking summary with movie title, date, time, seat numbers, and closing price

3. Navigation Path

- Movie details -> Book now -> Select showtime
Choose seats -> Confirm booking

The screenshot shows the 'Book Tickets' page. At the top, it says 'Online Movie Booking System'. Below that is a 'Title' section. Under 'Showtime', there is a dropdown menu labeled 'Select Showtime'. Below the dropdown is a 4x6 grid of 24 seats. The second row, third and fourth seats from the left are highlighted in red. Below the seat map, there is a 'Quantity' section with a 'Selected Seats' label and a 'Total Price' label. At the bottom, there is a red button labeled 'CONFIRM BOOKING'.

Register/Login

1. Input Elements

- Email and password fields
- "Register" or "Login" buttons
- Optional "Remember me" checkbox

2. Results Display

- Success confirmation message and redirection to homepage/user dashboard

3. Navigation Path

- Home -> Login/Register -> Dashboard

The screenshot shows the 'Register/Login' page. At the top, it says 'Online Movie Booking System'. Below that is a 'Title' section. Under 'Title', there are two buttons: 'Login' and 'Register'. Below the buttons is an 'Email' section with an input field labeled 'Email'. Below the email field is a 'Password' section with an input field labeled 'Password'. Below the password field, there is a checkbox labeled 'Remember me' and a link labeled 'Forgot password?'. Below these fields, there is a 'Quantity' section with a 'Selected Seats' label and a 'Total Price' label. At the bottom, there are two red buttons: 'LOGIN' and 'REGISTER'.

Cancel Booking

1. Input Elements

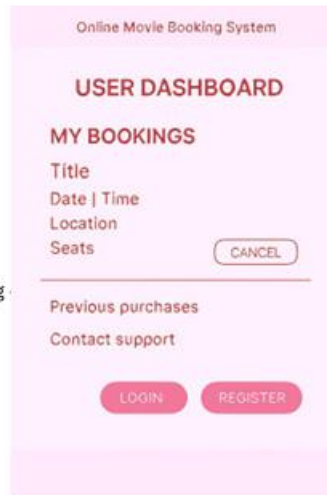
- User dashboard -> "My Bookings"
- Select booking -> "Cancel" button
- Confirmation pop-up

2. Results Display

- Cancellation confirmation message

3. Navigation Path

- Dashboard -> My Bookings -> Select Booking
- Cancel -> Confirmation



User Effort Estimation

Scenario A: Search and Book a Movie

1. Search for Movie:

- 1 click to focus search bar
- ~10 keystrokes to type movie name
- 2 clicks for filters
- 1 click to search

2. View Details:

- 1 click

3. Book Tickets:

- 1 click to select showtime
 - 3 clicks to choose seats
 - 1 click to confirm
4. **Total:** ~19 interactions (8 clicks, ~11 keystrokes)

User opens the Online Movie Booking System homepage. They click on the search bar at the top of the screen and type "Movie Title" using their keyboard. Then, they click the "Genre" dropdown and select "Sci-Fi." Next, they click the "Date" picker and choose "October 5." They click the "Location" dropdown and select "Gary Theater." After setting all filters, they click the pink "Search" button.

On the results page, they scroll through the movie posters and click the "View Details" button under "Movie Title." On the details page, they click the "Select Showtime" dropdown and choose "7:30 PM." Then, they click the "Book Now" button.

They are taken to the seat selection screen. They click three seats in Row C: C5, C6, and C7. Then, they click the "Confirm Booking" button. A summary page appears showing the movie title, date, time, selected seats, and total price.

Scenario B: Register and Cancel Booking

1. **Register:**
 - ~20 keystrokes (email + password)
 - 1 click to submit
2. **Cancel Booking:**
 - 2 clicks to navigate to booking
 - 1 click to cancel
 - 1 click to confirm
3. **Total:** ~25 interactions (5 clicks, ~20 keystrokes)

User clicks the "Register" button on the homepage. They click the "Email" text field and type "blah@example.com." Then, they click the "Password" text field and type "Password123." They check the "Remember Me" box and click the "REGISTER" button.

Later, they return to the site and click the "Login" button. They enter the same email and password, then click "LOGIN." They are redirected to their user dashboard.

The user logs into their account and is taken to the "User Dashboard." They scroll down to the "My Bookings" section. They see a booking for "Movie Title" on October 5 at 7:30 PM, Gary Theater, seats C5–C7.

They click the "Cancel" button next to the booking. A confirmation pop up box appears asking "Are you sure you want to cancel this booking?" They click "Yes." A message appears confirming the cancellation.

System Requirements

No.	Priority Weight (1-5)	Description
REQ1	1	Users can browse available movies and showtimes
REQ2	5	Users can book movie tickets online
REQ3	4	Users can cancel or modify bookings
REQ4	5	Admin and Agents can log in/out from the system
REQ5	3	Admin can add, update, or delete movie listings
REQ6	1	System sends booking confirmation via email
REQ7	2	Users can create and manage personal profiles
REQ8	1	System provides search and filter options for movies
REQ9	1	System logs user activity for audit and analytics

Use Cases

No.	Description
UC1	User browses movies and showtimes
UC2	User books tickets for a selected movie
UC3	User cancels or modifies an existing booking
UC4	Admin/Agent logs into the system
UC5	Admin manages movie listings (add/update/delete)
UC6	System sends booking confirmation to user
UC7	User creates or updates personal profile
UC8	User searches and filters movies based on genre, time, or location

Traceability Matrix

Req't	PW	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8
REQ1	5	X							
REQ2	2		X						
REQ3	5			X					
REQ4	4				X				
REQ5	2					X			
REQ6	1						X		
REQ7	1							X	
REQ8	1								X
REQ9	1								X
Max PW		5	2	5	4	2	1	1	1
Total PW		5	2	5	4	2	1	1	2